

A COUNTEREXAMPLE AT $n = 6$ TO THEOREM 1 AND CONJECTURE 36(3) OF KUNDU AND VELMURUGAN

ABSTRACT. We give a short calculation showing that

$$\text{ccn}(\chi_{(3,3)}; S_6) = \text{ccn}(\chi_{(2,2,2)}; S_6) = 5.$$

Thus the upper inequality in Conjecture 36(3) of Kundu and Velmurugan is false at $n = 6$; the same example also contradicts the bound in their Theorem 1, which excludes neither $(3, 3)$ nor $(2, 2, 2)$. The value itself is implicit in existing formulas for tensor powers of the irreducible representations of S_6 ; what is new here is the resulting contradiction with these two statements of Kundu and Velmurugan.

1. THE COUNTEREXAMPLE

For a character θ of a finite group G , let $c(\theta)$ denote its set of irreducible constituents. When it exists, define $\text{ccn}(\theta; G)$ to be the least positive integer k such that

$$c(\theta^k) = \text{Irr}(G),$$

where θ^k is the pointwise product, equivalently the character of the k -fold tensor power. Sun, Zhang, and Zhu computed power formulas for all irreducible representations of S_6 ; their formulas also imply the value below [3]. We give a short independent class-function proof and record its consequence for [1].

Theorem 1. *For the irreducible characters of S_6 indexed by $(3, 3)$ and $(2, 2, 2)$,*

$$\text{ccn}(\chi_{(3,3)}; S_6) = \text{ccn}(\chi_{(2,2,2)}; S_6) = 5.$$

Proof. Set $\chi = \chi_{(3,3)}$, and let π_r be the permutation character of S_6 acting on its r -subsets. Young's rule gives

$$\pi_3 = \chi_{(6)} + \chi_{(5,1)} + \chi_{(4,2)} + \chi_{(3,3)}, \quad \pi_2 = \chi_{(6)} + \chi_{(5,1)} + \chi_{(4,2)},$$

so $\chi = \pi_3 - \pi_2$. If C_μ is the conjugacy class of cycle type $\mu = (\mu_1, \dots, \mu_t)$, then

$$\pi_r(\mu) = [x^r] \prod_{i=1}^t (1 + x^{\mu_i}).$$

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Writing ε for the sign character, we obtain

μ	1^6	21^4	2^21^2	2^3	31^3	321	3^2	41^2	42	51	6
$ C_\mu $	1	15	45	15	40	120	40	90	90	144	120
$\chi(\mu)$	5	1	1	-3	-1	1	2	-1	-1	0	0
$\varepsilon(\mu)$	1	-1	1	-1	1	-1	1	-1	1	1	-1

The usual inner product of class functions therefore gives

$$\langle \chi, \mathbf{1} \rangle = \langle \chi^2, \varepsilon \rangle = \langle \chi^3, \mathbf{1} \rangle = \langle \chi^4, \varepsilon \rangle = 0.$$

For example,

$$\begin{aligned} 720 \langle \chi^4, \varepsilon \rangle &= 625 - 15 + 45 - 1215 + 40 - 120 \\ &\quad + 640 - 90 + 90 = 0. \end{aligned}$$

Hence none of $\chi, \chi^2, \chi^3, \chi^4$ contains every irreducible character of S_6 , and thus

$$\text{ccn}(\chi; S_6) \geq 5.$$

Miller proved that, for $n > 4$, every nonlinear $\psi \in \text{Irr}(S_n)$ satisfies

$$c(\psi^{n-1}) = \text{Irr}(S_n)$$

[2, Theorem 1.1(i)]. Since χ is nonlinear and $n - 1 = 5$ for $n = 6$, this gives $c(\chi^5) = \text{Irr}(S_6)$, and therefore

$$\text{ccn}(\chi_{(3,3)}; S_6) = 5.$$

Finally, $(3, 3)' = (2, 2, 2)$, so

$$\chi_{(2,2,2)} = \chi_{(3,3)} \varepsilon.$$

For every k ,

$$(\chi \varepsilon)^k = \chi^k \varepsilon^k.$$

Tensoring by the linear character ε^k permutes $\text{Irr}(S_6)$. Thus the two characters have the same covering number. $\square$

Corollary 2. *The upper inequality in Conjecture 36(3) of Kundu and Velmurugan [1] is false. The bound in their Theorem 1 is also false as stated.*

Proof. Conjecture 36(3) asserts, for even $n \geq 5$,

$$\text{ccn}(\chi_{(n/2, n/2)}; S_n) \leq \lceil \log_2 n \rceil + 1.$$

At $n = 6$, the right-hand side is 4, whereas Theorem 1 gives

$$\text{ccn}(\chi_{(3,3)}; S_6) = 5.$$

Their Theorem 1 asserts, for $n \geq 5$ and

$$\lambda \notin \{(n), (1^n), (n-1, 1), (2, 1^{n-2})\},$$

that

$$\text{ccn}(\chi_\lambda; S_n) \leq \left\lceil \frac{2(n-1)}{3} \right\rceil.$$

At $n = 6$, this bound is 4, and neither $(3, 3)$ nor $(2, 2, 2)$ is excluded. Theorem 1 therefore contradicts the stated bound. $\square$

Remark 3. The lower inequality in Conjecture 36(3) remains valid at $n = 6$. The counterexample does not address the corresponding upper bound restricted to even $n \geq 8$. In the proof of their Theorem 1, Kundu and Velmurugan state that the cases $n \leq 11$ were verified directly using SageMath; the examples above show that this finite verification cannot be correct as stated.

REFERENCES

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